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5 DECEMBER 1951
SUPERSEDING
Fed. Spec. FF–C–431
17 April 1942

FEDERAL SPECIFICATION

CLIPS, PAPER (BINDER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—Paper clips covered by this specification shall be designed for office use, to hold a specified thickness of paper.

1.2 Classification.

1.2.1 *Type.*—The paper clips shall be of the spring type, as specified herein.

1.2.2 *Classes.*—The paper clips shall be of the following classes, as specified:

Class 1.—With wire handles.

Class 2.—With die-stamped handles (Bulldog).

1.2.3 *Sizes.*—Unless otherwise specified, the sizes for class 1 and class 2 paper clips shall be in accordance with table I and table II, respectively, as specified.

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.)

2.2 Specifications and standard applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Materials.—Materials used in the manufacture of binder clips shall be first class and suitable for the purpose and be free from any defects which affect the appearance or which might affect the serviceability.

3.2 Design and construction.

3.2.1 *Class 1.*

3.2.1.1 *Components.*—Class 1 paper clips shall consist of a unit containing two hinged clip handles and one clip spring portion. The clip handles shall be detachable.

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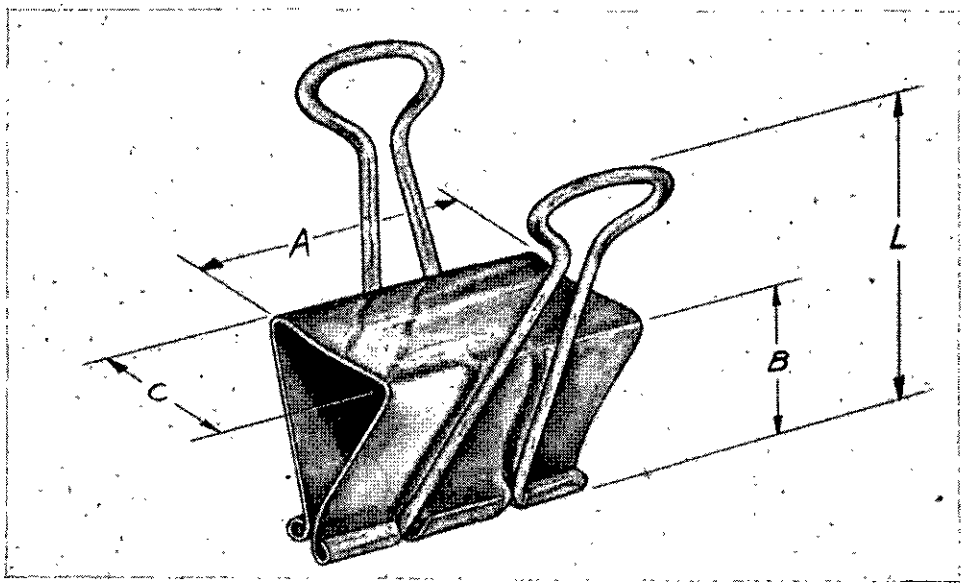


FIGURE 1.—*Class 1*

TABLE I.—*Dimensions of respective sizes, class 1 clips*

Size	Body or spring portion					Clip handle	
	A Over-all length	B Over-all depth	Thickness	C Over-all capacity	Capacity, maximum	L Over-all length, minimum	Diameter
	Dimensions in inches						
Large	2 ± 1/16	1¼ ± 1/32	0.022 ± 0.001	1 1/16 ± 1/16	1	2⅝	0.095 ± 0.005
Medium	1¼ ± 1/16	¾ ± 1/32	.016 ± .0005	⅝ ± 1/16	9/16	1½	.074 ± .001
Small	¾ ± 1/32	7/16 ± 1/32	.011 ± .0005	1 1/32 ± 1/32	9/32	1	.055 ± .001

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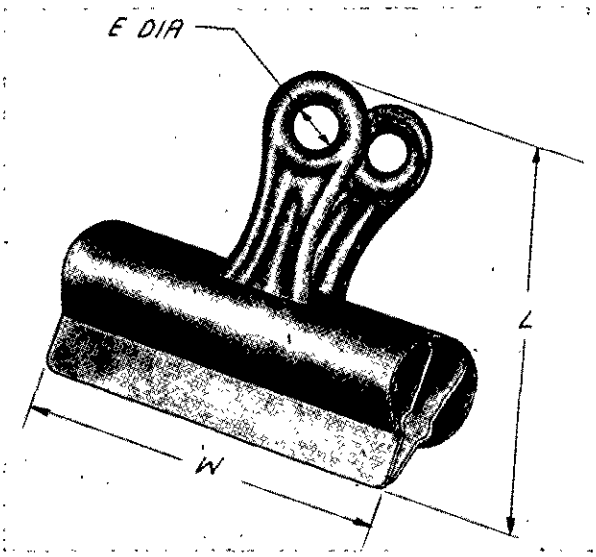


FIGURE 2.—*Class 2 (Bulldog).*

TABLE II.—*Dimensions of respective sizes, class 2 clips*

Size	W Width of jaw	E Diameter of eye, minimum	L Over-all length	Capacity, minimum
	Dimensions in inches			
Large	2⅞ ± ⅛	⅜	3 ± ⅛	1
Medium	2⅜ ± ⅛	⅜	2½ ± ⅛	½
Small	1 3/16 ± 1/16	7/32	2 ± ⅛	7/16

3.2.1.2 *Design.*—The general design of the class 1 paper clip shall be as indicated in figure 1. It shall be designed to maintain a uniform compression at all times, and shall meet the tests specified in section 4. The clip handles shall be so attached or hinged to the spring portion of the paper clip as to act as a device for opening the spring when compressed against the spring sides. The clip handles shall also serve as a means to hang the paper clip, or if required, the handles can be placed flat against the paper surface.

3.2.1.3 *Clip spring portion.*—The spring portion shall be made from clean, uniform, cold-rolled spring steel, having a chemical composition to produce a hardness that will give it the desirable qualities and enable it to meet the tests specified in section 4.

3.2.1.4 *Clip handles.*—Each clip handle shall be made of hard-drawn wire, suitable for the particular size clip. The clip handles shall retain their original rigidity and shape during normal use.

3.2.2 *Class 2.*

3.2.2.1 *Components.*—Class 2 paper clips shall consist of a unit containing two die-stamped handles and one cylindrical-shaped spring portion.

3.2.2.2 *Design.*—The general design of the class 2 paper clips shall be as indicated in figure 2. It shall be designed to maintain a uniform compression at all times and shall meet the tests specified in section 4. The handles shall be designed to have a countersunk eye in the upper portion of each blade and the lower portion shall be of sufficient width so as to serve as the

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clip jaws. The cylindrical spring shall be approximately equal to the width of the gripping edges of the jaws and shall be provided with two slits for the insertion of the handles. When assembled the jaws of the handles shall be in alignment one with the other and under the compression of the spring. The clip shall be so constructed as to prevent the jaws from becoming detached from the spring while in normal use.

3.2.2.3 *Clip spring portion.*—The spring portion shall be made from clean, uniform, cold-rolled spring steel, having a chemical composition to produce a hardness that will give it the desirable qualities and enable it to meet the tests specified in section 4.

3.2.2.4 *Clip handles.*—Each clip handle shall be of steel, malleable, properly annealed, and of a gauge suitable for the particular size clip. The clip hand grips shall retain their original rigidity and shape during normal use.

3.3 Finish.—All surfaces of the clip, including handles, shall be protected by a suitable finish which will not crack, chip or scale, and will leave no mark on paper. The surfaces of the clip shall be thoroughly cleaned and any dirt, oil, grease, and moisture shall be removed before applying the finish.

3.4 Marking.—Each clip spring body shall be stamped with the name of the manufacturer and trade-mark, if any.

3.5 Colors.—Unless otherwise specified by the procuring agency in the invitation for bids, colors normally in use for this type of item may be used.

3.6 Workmanship.—The binder clips shall be free from burrs, rough or sharp edges and any other undesirable surfaces. Dimensions and tolerances shall be as close as is consistent with the best shop practices and as indicated herein. Slight deviations in dimensions, resulting from normal manufacturing practices, will be permitted.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Not less than 5 clips of each size from each lot of 10 boxes or less, and not less than 10 clips of each size from lots in excess of 20 boxes, shall be taken at random for test purposes.

4.2 Inspection.—Samples shall be subject to such examination as may be necessary to determine compliance with the requirements of this specification.

4.3 Tests.—Tests shall be made in the order listed. Each sample failing to meet any one test shall be cause for rejection.

4.3.1 *Holding test.*

4.3.1.1 *Class 1.*—A sheet of paper having a thickness of not more than 0.005 inch shall be inserted between the jaws of each size class 1 binder clip and shall not slip in the jaws of the clip while holding a dead weight as follows: For the large size clip 16 ounces, for the medium size clip 12 ounces, and for the small size clip 6 ounces.

4.3.1.2 *Class 2.*—The holding test for class 2 clips shall be the same as specified in 4.3.1.1 except the dead weight shall be as follows: For the large size clip 16 ounces, for the medium size clip 16 ounces, and for the small size clip 12 ounces.

4.3.2 *Opening test.*

4.3.2.1 *Class 1.*—A pressure shall be placed on the ends of the handles which shall open the class 1 paper clips to approximately the half-way position; this pressure shall be 16 to 17 pounds for the large size, 14 to 15 pounds for the medium size, and 6 to 7 pounds for the small size.

4.3.2.2 *Class 2.*—The opening test for class 2 clips shall be the same as specified in 4.3.2.1 except that the pressure placed on the ends of the handles shall be 7 to 9 pounds for the large size, 5 to 7 pounds for the medium size, and 3 to 5 pounds for the small size.

4.3.3 *Capacity test.*

4.3.3.1 *Class 1.*—A block of wood $1\frac{1}{16}$, $\frac{5}{8}$, and $\frac{11}{32}$ inch thick, respectively, shall be inserted between the jaws of the large, medium, and small size class 1 paper clips and allowed to set for 24 hours. At the end of that time, each clip tested shall firmly hold a sheet of 32-pound sulphate bond paper. The spring shall not fracture at any point when opened to accept the block of wood.

4.3.3.2 *Class 2.*—The capacity test for class 2 clips shall be the same as specified in 4.3.3.1

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except that the paper clips shall have a block of wood 5/16, 1/4, and 1/4 inch thick, respectively, inserted between the jaws of the large, medium, and small size binder clips.

5. PREPARATION FOR DELIVERY

5.1 Unit packaging. — Unless otherwise specified in the invitation for bids, clips shall be packaged in cardboard or fiber boxes or containers containing 12 clips each. Packages shall contain clips of the same class and size.

5.2 Intermediate packaging.—Clips of one size only, packaged as specified in 5.1, shall be packaged in snug-fitting intermediate boxes such as folding cartons, set-up boxes, or fiberboard boxes of commercial grade and quality. The number of unit packages to be packed in an intermediate container shall be in accordance with best commercial practice.

5.3 Packing.—Clips of one size, packaged in unit and intermediate containers as specified in 5.1 and 5.2, shall be packed in corrugated or solid fiberboard boxes, style RSC, conforming to the requirements of Federal Specification LLL–B–631 or LLL–B–636. Gross weight shall not exceed 60 pounds. All flaps shall be securely sealed with a good quality adhesive applied throughout the entire area of contact between the flaps.

5.4 Marking.

5.4.1 *Issue packages.* — Cardboard boxes shall be marked with the item nomenclature, the class, size, quantity, and the name and trademark of the manufacturer.

5.4.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the item nomenclature, the class, size, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Intended use.—Paper binder clips covered by this specification are intended for use in offices, warehouses, shipping rooms, etc., to hold together sheets of paper conveniently, in definite order.

6.2 Ordering data.—Purchasers should specify the class and size of binder clips required and should exercise any desired options offered by this specification. (See 1.2.2, 1.2.3, tables I and II, 3.5, 5.1, 5.2, 5.4.2, and 6.3.)

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

MILITARY SPECIFICATIONS

JAN–P–103—Packaging and Packing for Overseas Shipment—Boxes; Wood-Cleated; Solid Fiberboard.

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

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MILITARY SPECIFICATIONS—Con.

JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[1]

MILITARY STANDARD

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Inspection procedures (Navy).*—For Naval purchases the general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.2.3 *Preparation for delivery.*

7.2.3.1 *Domestic shipment.*—Binder clips of one size shall be unit-packaged, packaged in intermediate containers, and packed in shipping containers as described in section 5.

7.2.3.2 *Overseas shipment.*—Paper clips of one size, packaged in unit and intermediate containers as specified in section 5, shall be packed in wood-cleated, solid-fiberboard boxes, wood-cleated, plywood boxes, or in Style 2 or 4, nailed wood boxes conforming to Military Specification JAN–P–103, JAN–P–105, or JAN–P–106. Only one type and style exterior container shall be used, gross weight not to exceed 150 pounds. Each shipping box shall be provided with a sealed case liner or sealed prefabricated bag of waterproof barrier-material, made in conformance with all the requirements of Military Specification JAN–P–125, type L–2 or M. Closure shall be completely sealed with a continuous seam at least ¾-inch wide of water-resistant adhesive complying with Military Specification JAN–P–140.

7.2.4 *Marking.*

7.2.4.1 *Unit and intermediate containers.*—Each box or carton shall be legibly labeled or stenciled to show the following information:

Stock No. _____
Fed. Spec. FF–C–431a
Nomenclature _____
Quantity _____
Size _____
Name and trade-mark of manufacturer____

7.2.4.2 *Shipping containers.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of Military Standard MIL–STD–129.

7.2.5 *Ordering data.*—Requests, requisitions, schedules, and contracts or orders should specify whether domestic or overseas packing and marking are required. (See 5.3, 5.4, 7.2.3, and 7.2.4.)

NAVY INTEREST: *MC* S

[1] Applicable to Navy purchases only.

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog

For sale by the Superintendent of Documents, Washington 25, D. C. Price 5 cents.

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